

News Bias Detection during Palestine-Israel Conflict

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Abstract

This study investigates linguistic bias in international reporting of news of the Palestine-Israel conflict across 18 prominent English-language newspapers. It utilises advanced Natural Language Processing techniques like semantic role labelling, Named Entity Recognition and coreference resolution to systematically examine the agents and patients in violent acts being reported. The analysis reveals consistent patterns of asymmetric agency attribution across various media outlets. The findings highlight the impact that syntactic packaging and lexical choices have on the narrative that is created around the conflict. A region-based analysis also proves the Indexing Theory in the case of the United States, United Kingdom and India. It takes established linguistic theories and scales it with computational methods to conduct a deep analysis.

Research Ethics Approval

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This study does not involve any human participants.

Declaration

I declare that this thesis was composed by myself, that the work contained herein is my own except where explicitly stated otherwise in the text, and that this work has not been submitted for any other degree or professional qualification except as specified.

(Krishi Naresh Jain)

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Chapter 1

Introduction

This chapter introduces the motivation behind creating a systematic method of analysing the reporting of violent events and analysing the media bias during Palestinian-Israeli conflict. It discusses the structure of the dissertation and the research questions that it aims to answer.

1.1 Motivation

The portrayal of violent conflicts in news media is a powerful force shaping public understanding and political discourse. Language choices, like how agency and victimhood are linguistically constructed, influence readers' perceptions of responsibility, legitimacy, and morality. In highly contested and polarized conflicts such as the Israeli-Palestinian one, news framing can often reinforce narratives and impact international opinion. Advances in natural language processing (NLP), including semantic role labeling, Named Entity recognition, coreference resolution and verb classification, enable large-scale, systematic investigations into these subtle linguistic mechanisms and the underlying media bias.

On October 7th, 2023, members of Hamas launched an attack on Israel, killing nearly 1,200 people and taking more than 250 hostages (AP News) [6]. Israel followed by launching multiple attack campaigns into Palestinian territories and has continued to do so until present-day. Various outlets have been covering these events and reporting on them. Media coverage of this conflict is important for presenting facts and information, and also framing the narratives for their audiences, globally.

This coverage across outlets has since been often called out for bias. The Asserson Report [28], published in September, 2024, claimed that the BBC was biased against Israel in its reporting of the events. According to this report, the BBC was omitting facts, reporting inaccurately, was unfair in its coverage of casualties and used ambiguous language. Following this, articles like openDemocracy’s [17] *Media research shows BBC is very far from ‘biased against Israel’* contradicted these claims.

This subsequently makes it more difficult and confusing for readers to decide which sources to trust. They need a way to form informed opinions and be able to ensure that the information they have is complete and reliable. This highlights the need for a systematic way to quantify the bias that exists, if any.

1.2 Research Questions

The study is guided by the following research questions:

- **How are agents and patients of violence linguistically encoded across different news sources covering the Israeli-Palestinian conflict?** - This question explores whether certain actors are systematically foregrounded or backgrounded, and how their roles as perpetrators or victims are represented through argument structure and verb usage. It also considers if the region of the newspaper affects how they present information.
- **How effectively can computational methods such as semantic role labeling combined with verb classification capture and quantify these framing phenomena?** - This investigates the reliability and validity of NLP approaches to analyze complex media discourse, providing a replicable methodology for media bias analysis.

By answering these questions, the study aims to contribute novel insights into the language of conflict reporting and the role of syntactic-semantic structures in shaping public narratives of violence.

1.3 Objectives

The primary objective of this study is to investigate the presence of linguistic bias in the reporting of the Palestine-Israel conflict in some of the most popular English-language

newspapers around the world.

1. **Data Collection** - Build a way to scrape data from 9 different sources and create a corpus of news articles for further analysis.
2. **Semi-Supervised Classification** - Train a highly effective classification model to filter out irrelevant articles and reduce noise from data.
3. **Semantic and Syntax Based Analysis of Violence Reporting** - Use computational linguistic techniques and Natural Language Processing to label semantic roles in sentences and analyse the shaping of narratives of violence.

1.4 Thesis Structure

This thesis is structured as follows:

- **Chapter 2** gives some context to the conflict and critically analyses existing literature on media bias and the use of different Natural Language Processing techniques for bias detection.
- **Chapter 3** discusses the data collection and creation of the corpus. It also explains the pipeline built for semi-supervised classification of articles based on their relevance. It analyses the number of articles posted by different news sources over the given time-frame and compares them with significant events that have taken place regarding the conflict.
- **Chapter 4** presents the methods used to identify the framing bias in the articles. It discusses the results observed and the interpretation of these results and analyses what the results imply.
- **Chapter 5** summarises the contributions of the research, its limitations and the future work that can be done to improve and help us contextualise and quantify media bias in regards to this conflict.
- **Appendix A** contains details on where the project materials can be accessed.

Chapter 2

Background

This chapter focuses on building context around the conflict and discusses previously conducted research in related areas that form the foundation of the dissertation. It identifies the gaps that exist in current literature and positions this study and its necessity.

2.1 The Palestine-Israel Conflict

In 1917, the Balfour declaration expressed support for “the establishment in Palestine of a national home for the Jewish people.” The League of Nations put Palestine under the administration of the UK in 1922 in a mandate over Palestine. Following this, the Arabs demanded freedom and resisted Jewish immigration, leading to a rebellion in 1937. In 1947, the UK handed over this issue to the United Nations which considered all its options and partitioned Palestine into two parts. Multiple clashes broke out between the Jews and Arabs in Palestine [?]. In early 1949, Israel had occupied most of the region, including the Egypt-Palestine frontier, except for the Gaza Strip. Israel recognises this as their War of Independence while the Arabs know it as Nakba (meaning ‘the catastrophe’). This was because it resulted in a large number of refugees being displaced as a result of this war [4]. Since then, multiple wars have broken out between Israel and the surrounding Arab nations (Egypt, Jordan, Syria, Iraq and Lebanon) [1]. The United Nations Security Council adopted the United Nations Resolution 242 in November 1967 which called for the Arab states to accept Israel’s right “to live in peace within secure and recognized boundaries free from threats or acts of force.” After Hamas’s terror attack on October 7th, 2023, Israel started its counter-attack. Since then, the developments in this conflict have been reported on by news media around the world, which is the focus of this dissertation.

2.2 Literature Review

2.2.1 Media Bias

Media bias across different geo-political conflicts has been a focus of many studies in journalism, political science and computational media analysis. A common challenge is that bias is highly subjective; a reader's cognitive and cultural biases make it difficult to accurately recognise bias when consuming news. For conflicts with a long history, it is more difficult for the audience to objectively tell whether a news article they read is biased or not. Their personal opinion can often cloud their perception. It becomes even more important for news media to present facts with as little bias as they can. According to Mullainathan and Shleifer [22], bias can be of two kinds - ideological and spin bias. Spin bias is when news is 'created' to be published, often sensationalizing non-issues. Ideological bias can happen in two ways: when the outlet has a particular ideology that causes them to manipulate the news they report, and when they spin the news to cater to their readers themselves. If we talk about competing newspapers, ideological manipulation has a net zero effect, but their audiences' beliefs still affects the paper's bias itself. This happens because outlets believe that future newspapers, covering these same issues, will reinforce readers' beliefs. This pressures the media source to cater to readers' existing biases. Which brings up - How is a reader supposed to recognise if the news they consume is presenting facts or if it is catering to their beliefs, reinforcing them?

Building on this, agenda-setting and framing theories conceptualize how media shape not only what people think about but also how they interpret information [21]. Agenda-framing tells people what they should think about and how they should think about it. News coverage of international politics greatly influences the public opinion on these issues. Framing is the selection of some aspects of perceived reality and making it more noticeable so as to promote a particular problem definition, causal interpretation, moral evaluation, and/or treatment recommendation for the item described' [26]. These studies talk about individual aspects of recognising different ways in which bias can be present in the media. However, the nuance of these requires time and an unbiased evaluator to properly identify, especially at a large scale. This is where computational techniques come in.

2.2.2 Using Natural Language Processing in Linguistic Bias Detection

In his 1998 paper, Van Dijk [29] talks about the roles of semantic content ("event models") and linguistic form ("context models") in constructing biased narratives. [14] details the various ways in which the language used in news can affect the views of their readers with subtle linguistic mechanisms: voice, modality, etc. The study also talks about how linguistic bias is not restricted to lexical labeling but extends to syntactic-semantic structures. Both these studies have shown that agents of violence are sometimes syntactically hidden or deflected through passive constructions, while victims are foregrounded via patient roles, thereby influencing the perceived responsibility of the violent act itself. More recently, it has been noted that syntactic packaging can convey implicit sentiment independent of the words used explicitly in a sentence [27]. This gives us a crucial insight in comprehending how agency and blame are subtly encoded in conflict reporting. Their findings reveal that even in the absence of overt sentiment markers, syntactic structures such as voice, argument omission, or modifier placement serve to implicitly guide readers' emotional interpretations. This insight is pertinent to media reporting on violence: subtle grammatical choices can modulate the perceived agency and blame directed at conflicting parties without explicitly stating the same.

Computational techniques have advanced bias detection, particularly with the rise of semantic role labeling (SRL), which automates extraction of agent-patient relations and modifiers, enabling extensive mapping of "who does what to whom" [16] [24]. Other NLP approaches utilize sentiment analysis, word-choice labeling [18], and transformer-based classifiers fine-tuned on polarizing datasets to enhance bias recognition accuracy. These used a bias recognition module to annotate articles as biased or not. Raza et. al. [25] made the model look for bias-bearing words associated with each entity in the Named Entity Recognition (NER) task. Ness, et. al. [23] trained BERT, logistic regression, random forest, multinomial, and long short-term memory (LSTM) on highly polarising news sources. This study also showed that training models on these sources improve bias detection accuracy. They divided their news sources into three categories: mainstream media sources, known conservative news sources and known liberal news sources. The political standing of sources proved to influence the bias they display.

2.3 Bias Detection in Israel-Palestine Conflict

Coverage of the Israeli-Palestinian conflict in global news media is frequently subject to intense scrutiny concerning bias, ideological framing, and selective representation. Kozman and Cozma [19] analysed the headlines of articles published by the Associated Press (AP) and Qatar News Agency (QNA) based on the rationale that the headline of a story is the most salient framing feature of it and that a large portion of the audience only surfs through the headlines to get their information. This was done by using measures of war journalism and peace journalism. The results showed that even though QNA did more peace journalism, they also demonised Israel more in the language they used and victimised and advocated for Gaza more. AP, on the other hand, did less labelling on either side. It also acknowledges that the cultural and geo-political proximity to the situation influences how either news outlet covers the conflict. Hamborg et. al [18] introduce word-choice labelling as an automated way of identifying media bias. This refers to when different articles refer to the same concepts or events using different terms to influence reader perception.

In 2020, Friedman and Herfroy-Mischler [15] analysed the difference in blame agency among Israeli, Palestinian and American newspapers for the 2014 conflict. The blame occurrence was defined as "an agent accusing another agent of responsibility for bringing about a negative situation". Two manual coders selected blame occurrences and the nationality of the blaming and blamed agents were assigned. While this is a solid foundational method, scaling it to more than six newspapers and over a longer period of time with human annotation is infeasible. This paper [10] analysed 13 articles over 12 days, aired on CNN and found that the majority of the conflict coverage focused on the Israeli perspective. This paper is also limited in its analysis in terms of number of articles analysed and the date-range covered. The use of active and passive voice when reporting events based on the entity performing the action also showcases some bias in who is shown to have agency. The study also analysed the usage of violent verbs like 'killed', 'murdered' and 'died', along with the voice used to report these acts.

Based on the analysis of previously conducted related research on specifically the topic of the Palestine/Israel conflict, they are limited in terms of their diversity of opinions, the dataset size or the use of a small number of models. This study seeks to address these limitations by combining established linguistic analysis techniques with scalable approaches that can effectively handle larger and more diverse corpora.

Chapter 3

Data

This chapter discusses the collection of data from different sources, the considerations made for different outlets and the additional data augmented to create a richer corpora. It then defines the methods used to classify articles as relevant to the study or not. Finally, it discusses the patterns observed in the frequency of articles published by different outlets.

3.1 Data Sources

The dataset for this study was compiled from a curated list of prominent news websites. The selection process began with the Press Gazette's 50 Most Popular English-Language News Websites in the World (April 2025) [8]. From this list, only websites without paywalls were considered for data collection, and their respective robots.txt files were reviewed to ensure compliance with web crawling guidelines.

Ultimately, nine websites were selected for direct scraping: ABC News, AP News, CNBC News, Daily Mail UK, Fox News, The Express, The Independent UK and The Guardian. In addition to these, data from ten further sources, that were collected by a colleague for a related dissertation, were augmented into the final dataset.

3.2 Data Collection

All the articles analysed in this study were collected using Web Scraping. Web scraping is the process of extracting data from different websites (in this case, websites for different news outlets) using automation [7].

3.2.1 Implementation

Data was gathered using a hybrid approach that combined Scrapy spiders for large-scale crawling and BeautifulSoup4 for HTML parsing. According to the Scrapy documentation [5], "Scrapy is a fast high-level web crawling and web scraping framework, used to crawl websites and extract structured data from their pages." The BeautifulSoup4 documentation [2] describes it as: "Beautiful Soup is a library that makes it easy to scrape information from web pages. It sits atop an HTML or XML parser, providing Pythonic idioms for iterating, searching, and modifying the parse tree."

A spider script describing how to traverse and store articles was defined for each news source:

ABC News - ABC News has sitemaps as HTML pages for each month, divided into parts of 500 articles per page. The spider was scripted to read all 500 links before moving onto the next for any given month between October 2023 and May 2025.

AP News - AP News has XML sitemaps for each month. This sitemap was traversed and all articles were subsequently visited.

CNBC - All articles ever published to their website are stored in large XML sitemaps. For articles relevant to the study, Page 12 was specifically scraped.

DailyMail UK - The robots.txt was recently updated to disallow Scrapy spiders, so the crawler was designed to not use a spider and only get the data using requests and BeautifulSoup4.

Fox News - Fox News also has sitemaps divided into pages for every article ever published together. So, pages 1-12 at the time were scraped.

The Express - BeautifulSoup4 along with Python requests library was used to traverse month-wise XML sitemaps.

The Independent UK - A Scrapy spider was used to traverse sitemaps and scrape articles.

The Guardian UK - The Guardian has topic pages defined. All articles in the /gaza, /israel and /middleeast topics were traversed.

The Sun UK - The script for this used Scrapy and BeautifulSoup4.

Scrapy handled URL discovery, pagination, and sitemap traversal, while BeautifulSoup4 was used to extract structured article elements (e.g., title, publication date, body text) from raw HTML. After this, for each article URL that was visited, a decision

was made on whether to scrape it or not, based on the presence or absence of certain keywords.

3.2.2 Keyword-Based Filtering

To be scraped, an article must contain at least one of the predefined keywords as defined in [12].

The keywords were (case insensitive): Palestine, Israel, Palestinian/s, Israeli/s, Gaza, Hamas, IDF, Israeli Defence Force/s, Israeli Defense Force/s, and variations of the date October 7th.

Limitations: This filtering technique used covered general keywords related to the conflict. However, there could have been instances where none of these words were mentioned but the article was still relevant to the topic. Another potential issue was the scraping of irrelevant articles (like October 7th causing unrelated articles to be scraped). This was addressed by using machine learning models of classification, as discussed in 3.4.

The total number of articles collected were 115778 across 19 different sources, individual numbers can be found in 3.2.

3.2.3 Date-Based Filtering

A lot of articles were scraped from month-based sitemaps. To filter out articles from outside of our defined date-range, a second filter was added to make sure the articles were published between 7 October 2023 and 31 May 2025 (inclusive).

3.3 Data Pre-Processing and Filtering

After the articles were scraped, they were cleaned to remove unnecessary HTML, whitespaces and irrelevant elements (e.g., navigation menus, footers, advertisements) that might have been scraped. Character encoding was standardized to UTF-8, and duplicates were removed based on matching URLs and text similarity. Each article was given a unique ID.

AP News - Articles scraped from the AP News website were in two languages, English and Spanish. Using Python library langdetect, articles were classified based on this and only English articles were considered for the study.

The Sun UK - The spider was only able to collect articles from November 2024 onwards. Thus the source was excluded from further analysis to maintain uniformity.

3.4 Data Classification

3.4.1 Relevance of Articles

For the purpose of the study, 4 different categories were defined:

1. Articles directly reporting on the conflict.
2. Articles reporting on protests or public reactions to the conflict.
3. Articles reporting on statements made by political/official figures addressing the conflict.
4. Articles that were scraped due to a keyword being present but not necessarily about the conflict.

Of these, categories 1 and 3 were considered directly relevant to this study.

3.4.2 Data Labeling

A random sampling of 200 articles were chosen. Two annotators classified the articles as relevant and irrelevant on the basis of the above directions. The results of their classification are shown in 3.1.

There was a clear agreement on category 2 and 4 of articles. For categories 1 and 3, there was a 4% disagreement and 34% disagreement respectively. Articles where the annotators could not agree, a third annotator was asked to label the articles as relevant or irrelevant and to classify them.

3.4.3 Supervised Classification

Four different classifiers were trained on the labelled dataset - Logistic Regressor, Random Forest, Support Vector Machine (SVM) and DistilBERT. They were trained and evaluated using an 80/20 split and assessed with a 5 fold cross-validation. The results are summarised in 3.1. SVM and DistilBERT showed similar high accuracy, but SVM's recall and F1 scores lagged behind. Logistic Regression had a very high

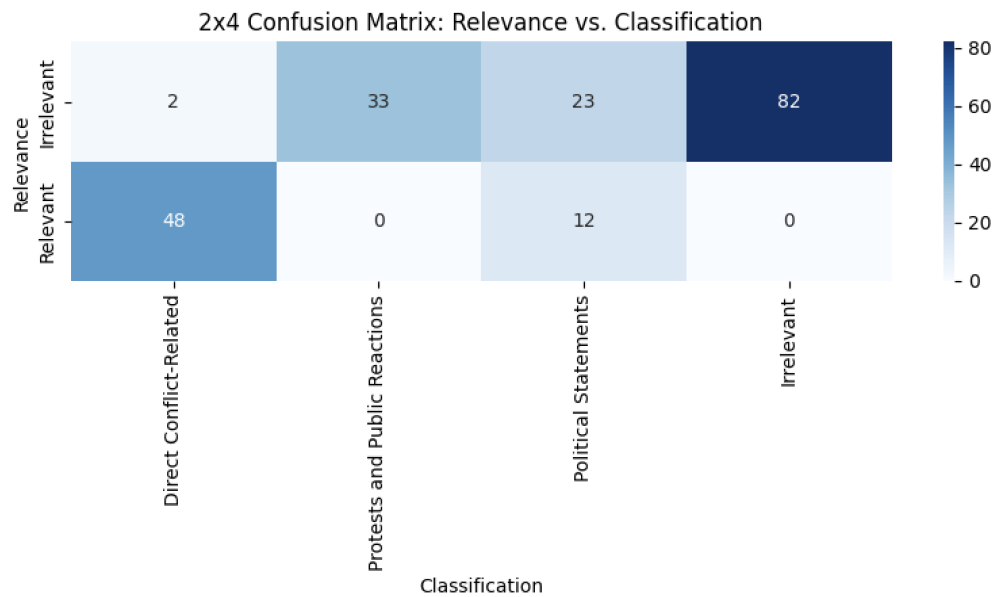


Figure 3.1: Relevance v/s Classification of 200 Articles

precision of 0.91 but its recall fell to 0.69, which meant it was missing relevant articles. To address these limitations, a semi-supervised approach was built on top of the best-performing from these models, DistilBERT.

3.4.4 Semi-Supervised Classification

A semi-supervised active learning approach was adopted to effectively address the challenge of limited labeled data for text classification. These class labels were used to train a DistilBERT-based classifier. To leverage a larger unlabeled corpus, the trained model was applied to predict labels and corresponding confidence scores on the unlabeled data. 100 articles with low-confidence predictions were selected and manually reviewed to provide accurate annotations, to improve the quality and representativeness of the labeled dataset. Simultaneously, 52 articles with high-confidence predictions were pseudo-labeled and added to the training set to augment data volume and help balance class distribution to use self-training [30]. The combined dataset of manually labeled and pseudo-labeled articles [20] was then used to retrain the classifier. This iterative process of selective manual annotation and pseudo-labeling allowed for progressive enhancement of the model's performance, striking a balance between label accuracy and dataset scale.

This showed significant improvement in the models' performances. Since DistilBERT

still performed better than all other models in most metrics, this trained classifier was used on the entire dataset again. Table 3.2 shows the final counts of the number of articles classified as relevant.

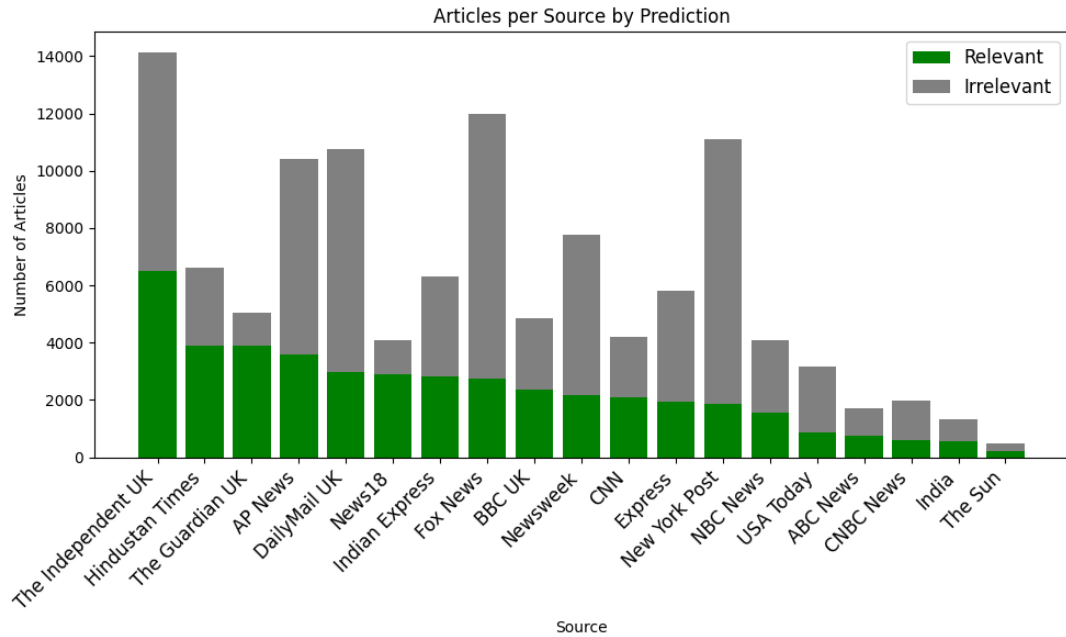


Figure 3.2: Relevant articles after filtering and classification.

Method	Model	Accuracy	Precision	Recall	F1
Supervised	Logistic Regression	0.84	0.91	0.69	0.78
	Random Forest	0.82	0.83	0.74	0.77
	SVM (LinearSVC)	0.88	0.90	0.74	0.77
	DistilBERT	0.88	0.85	0.88	0.86
Semi-supervised	Logistic Regression	0.88	0.90	0.87	0.88
	Random Forest	0.87	0.85	0.90	0.87
	SVM (LinearSVC)	0.89	0.91	0.87	0.89
	DistilBERT	0.90	0.87	0.93	0.90

Table 3.1: Performance evaluation of different models under semi-supervised and supervised settings

News Source	Articles Scraped	Articles After Filtering and Classification
ABC News*	1705	747
AP News*	10416	3584
BBC News UK	4848	2378
CNBC*	1958	582
CNN	4190	2085
DailyMail UK*	10746	2990
Express UK*	5805	1919
Fox News*	11969	2749
Hindustan Times	6619	3889
India	1326	539
Indian Express	6291	2818
NBC News	4100	1566
News 18	4098	2887
Newsweek	7778	2163
New York Post	11106	1877
The Guardian*	5028	3884
The Independent UK*	14137	6511
The Sun*	498	207
USA Today	3160	877
Total	115778	44251

Table 3.2: Number of articles collected and number of articles after cleaning, filtering and classification. * indicates news sources collected in the scope of this project.

Chapter 4

Bias Analysis

This chapter discusses the methodologies used to perform analysis on the data collected. For this bias analysis, NLP techniques like Named Entity Recognition, Semantic Role Labelling and Coreference were used.

4.1 Data Preparation

4.1.1 Sentence Segmentation

The first step was to divide all the articles into sentences. This was done to facilitate a deeper analysis in the language used. Each sentence was given a weight based on where it occurs in the article:

1. The headline
2. The description (if the article has one)
3. Leading paragraph (the first three sentences of the body)
4. Body

This was done using the `nltk.tokenize` library.

4.1.2 Semantic Role Labelling - Verb Detection

Each sentence was further processed using Semantic Role Labeling (SRL). AllenNLP's SRL models (based on PropBank-style SRL [11]) predict predicate–argument structures around verbs. The tags follow a BIO scheme (Begin–Inside–Outside), meaning each

argument is marked with B- (begin), I- (inside), or O (outside / not part of any role). Some of the resulting SRL labels are explained in 4.1. The primary focus was on ARG0 (typically the agent/initiator of the action) and ARG1 (typically the patient/recipient of the action). While SRL can provide candidates for agents and patients, there is often a lack of explicit identification, the use of pronouns, etc. To resolve this, NER and Coreference Resolution were used.

Tag	Meaning
ARG0	Proto-agent (doer, experiencer, initiator)
ARG1	Proto-patient (thing affected, acted upon)
ARGM-LOC	Location
V	The predicate verb itself
B- / I- prefix	BIO encoding: Begin / Inside of a span
O	Outside (not part of any role)

Table 4.1: Semantic Role Labeling (SRL) Tags in AllenNLP Used (PropBank style [11], BIO encoded)

4.1.3 Extracting Agent-Patient Information using Location

In addition to identifying agents and patients, extracting pertinent location information was approached by leveraging the location argument (ARG-LOC) provided by the SRL model. ARG-LOC captures spatial details associated with the verb and its participants, such as the place where an action occurred or where an agent or patient was situated. By systematically collecting ARG-LOC annotations, it was possible to associate violent events with specific geographic contexts—critical for analyses regarding region-specific impacts or framing. For instance, in the sentence “you’re murdering people in Gaza,” the phrase “in Gaza” is encoded as ARG-LOC, linking the patient (“people”) to a specific location. Incorporating ARG-LOC into the event representation thus offers richer, context-sensitive descriptions and enables downstream analyses on the geography of violence depicted in news reports.

4.1.4 Named Entity Recognition

Named Entity Recognition (NER) is the automated identification of named entities (such as people, organizations, locations) within text. Each ARG span was passed

through a transformer-based NER model (spaCy/transformers) to detect whether it contained references to relevant actors (e.g., “Israel,” “Palestinian militants,” “IDF,” “Hamas”). A mapping dictionary consolidated entity variants (e.g., “Israeli forces” → Israel). The output of NER would look like:

Sentence: “The Israeli military confirmed that it had targeted al-Ahmad, saying he was a commander with Hamas in south Lebanon and was behind several attacks against Israel.” (AP News, 7th May 2025) [3]

Entities: [“Israeli”, “NORP”], [“al-Ahmad”, “PERSON”], [“Hamas”, “ORG”], [“Lebanon”, “GPE”], [“Israel”, “GPE”]]

4.1.5 Coreference

Accurate extraction of agent-patient relationships in complex sentences often requires resolving coreference, where entities are referred to indirectly through pronouns or noun phrases. In the annotation and evaluation process, coreference resolution algorithms were utilized to link pronouns (e.g., “he,” “they,” “his troops”) or definite descriptions back to their corresponding entities in the text. This step was crucial for handling instances where the agent or patient was distributed across multiple sentences or clauses. For the given example, the initial sentence introduced “Israelis” and subsequent sentence referred simply to “you’re”, coreference resolution enabled the system to maintain continuity of entity tracking and assign agent or patient roles accurately. This process involved integrating SRL outputs with coreference chains, thereby enriching semantic analysis and reducing mislabeling resulting from ambiguous pronoun usage. The `fastcoref` library was used for this.

4.1.6 Filtering Violent Sentences

In order to restrict the analysis to events involving violence or fatalities, a systematic filtering procedure was implemented. All verbs detected in the previous preprocessing phase were first lemmatized to their base forms, ensuring consistent comparison regardless of verb conjugation or tense. To identify violence-related predicates, VerbNet v3.2—a comprehensive lexical resource that organizes English verbs into semantic classes—was employed. Specific VerbNet classes and entries associated with violent action and fatalities (such as “kill,” “attack,” “bomb,” “massacre,” etc.) were curated into a violence verb list. Each sentence in the dataset was inspected for the presence of these violent verb lemmas. Only sentences containing at least one of the designated verbs

were retained for further semantic analysis, thus focusing on agency representation within violent events.

4.2 Agent-Patient Detection

To determine the assignment of agency within violent sentences, a pipeline applying SRL, NER and coreference resolution was built.

The categorization of agents and patients was defined as follows:

1. Israeli (I)
2. Palestinian (P)
3. Unknown or Unclassified (U)

4.2.1 Assigning Israeli-Palestinian Classes

Each sentence was labelled with its semantic roles and sentences containing violent verbs were selected. These sentences were then processed with coreference in the context of the entire article (using the unique article IDs). For each violent verb instance, If ARG0 and ARG1 could not be directly linked to a side using NER, ARGM-LOC (location modifiers) were used as secondary evidence. For instance, “fighters in Gaza launched rockets” would be attributed to the Palestinian side due to the locative marker. When both NER and ARGM-LOC failed to provide disambiguation, the argument was marked as Unknown. and the agent and patient for each verb were generated. These lists of agents, patients and locations were then labelled as Israeli or Palestinian. A few examples of the terms used to classify are available in Table 4.2

	Israel	Palestine
Terms	'Israeli', 'IDF', 'Israeli Army'	'Palestinian', ' Hamas', 'Gazans'
Location	'Tel-Aviv', 'Israel'	'Palestine', 'Gaza', 'West Bank', 'Rafah', 'Khan Younis'

Table 4.2: Examples of terms used to assign classes to agents and patients

Limitations: This method has a few limitations which might cause misclassifications of the agents and patients.

- Less common words might have been overlooked. In cases where there is usage of a particular location term that was not identified in either classes, it will be automatically counted as an Unknown class.
- Indirect references in sentences with no explicit indication of which side they belong to will be misclassified. For example, terms like 'the troops', 'the terrorists'.
- The static nature of the term lists requires constant updating to incorporate emerging terminology, new factions, or evolving language use in news reporting, demanding ongoing maintenance of the classification lexicon.

4.2.2 Classification Evaluation

A random sample of 60 sentences containing violent verbs (as identified by the value of V) was selected for manual annotation. To make this sample size balanced, 3-4 sentences from each source were chosen. A human annotator assigned agent and patient categories to each verb instance from the classes above. The results of performance evaluation tests run are in Table 4.3. With a sample size of 60, the accuracy estimates carry a margin of error of about $\pm 11\%$. Considering the huge difference in the sample size and the actual number of articles, the percentages may fluctuate noticeably with a different sample. With this evaluation, the model was best at agent-identification. There were no observed significant differences in error rates across outlets, although the sample size is too small to rule out systematic variation. A larger, stratified evaluation would be needed to confirm whether the model struggles more with particular sources.

Task	Accuracy
Agent Identification	83.33%
Patient Identification	78.33%
Agent-Patient Set Identification	71.66%

Table 4.3: Evaluation of enhanced agent-patient detection using SRL and Coreference

The different test sentences that the model's accuracy statistics are based on, are elaborated upon in Table ???. The errors were primarily attributed to ambiguous phrasing or long sentences that proved difficult for the coreference model to follow through on. The details for the occurrence of different kinds of errors are discussed in Table 4.4.

Error Type	Count	Percentage
Vague words (e.g., 'the troops', 'people')	9	52.9%
Long / complex sentences	5	29.4%
Unknown locations	3	17.6%
Total	17	100%

Table 4.4: Distribution of the 17 agent–patient identification errors by error type.

Source	Sentence	Predicted Agent	Predicted Patient	Analysis	Result
ABC News	The IDF citing "the assessment of the professional authorities" said Ariel and Kfir Bibas "were brutally murdered in captivity in November 2023 by terrorists."	Israel	Israel	In this sentence, the term 'by terrorists' does not explicitly mention who it could be referring to. Using coreference, the model assigned it to Israel, even though the blame for the verb 'murdered' was attributed to Hamas.	Incorrectly identified Agent
ABC News	The IDF also confirms it struck Nasser Hospital in Khan Younis, claiming it killed multiple members of Hamas.	Israel	Palestinians	The model correctly identifies the agent and patient for the verb "killed" using coreference (associating 'it' with 'The IDF' and using ARG-LOC mapping from Khan Younis).	Both correctly identified.
AP News	In the Oct. 7 attack, militants killed about 1,200 people, mostly civilians, and abducted 251.	Unknown	Unknown	The lack of an explicit keyword specifying the agent or patient makes it difficult for the model to categorise "the militants" and "people, mostly civilians".	Both Agent and Patient Incorrectly Identified
CNN	"Only later, after approaching the vehicles and scanning them, was it discovered that these were indeed rescue teams." The IDF said that six of those killed were identified "in a retrospective examination as Hamas terrorists," raising the question of whether the troops opened fire on ambulances before knowing if there were any militants amongst the vehicles."	Unknown	Unknown	The long and complicated sentence makes it difficult for the coreference model to track agents and patients. Additionally, "the troops" make it ambiguous.	Incorrectly Identified Agent
CNBC	Earlier Israeli airstrikes killed at least 15 people, eight of them in the Rafah area, said Palestinian health officials.	Israel	Unknown	The model failed to track and associate "in the Rafah" area to the "15 people".	Incorrectly Identified Patient

Source	Sentence	Predicted Agent	Predicted Patient	Analysis	Result
DailyMail UK	Islamic Jihad in Gaza fires two rockets towards Israel after Israeli troops raid a refugee camp and kill seven Palestinian gunmen and two civilians.	Israel	Unknown	For the verb 'raid', while the agent is correctly assigned, the lack of directly mentioning where the refugee camp is makes it assign the patient as Unknown.	Incorrectly Identified Patient
Fox News	The military seized a strategic corridor along the border and the Rafah crossing with Egypt, which is Gaza's only path to the outside world that was not controlled by Israel.	Israel	Unknown	While coreference identifies the agent for the verb 'seized' correctly, the location being the border creates ambiguity on which side to assign the Patient side as.	Incorrectly Identified Patient
Indian Express	One Israeli soldier was also killed in clashes.	Unknown	Israel	The model correctly identifies the victim and the lack of the agent of violence.	Both correctly identified.
The Guardian	Israeli forces kill Hamas gunmen in raid near West Bank's Tulkarm.	Israel	Palestinians	The model correctly picks the agent and patient in this sentence.	Both correctly identified.
USA Today	Dozens in Gaza killed at post office, near aid trucks, as US renews push for cease-fire	Unknown	Palestinians	The patient is identified correctly and the sentence does not contain any information on how the victims were "killed", so the agent is tagged as Unknown.	Both correctly identified.
The Independent UK	Israeli strikes have killed more than 54,000 Gaza residents, mostly women and children, according to the Gaza Health Ministry, which does not distinguish between civilians and combatants in its tally.	Israel	Palestinians	ARG0 - "Israeli strikes" and ARG1 - "54,000 Gaza residents" are processed with SRL.	Both correctly identified.
Indian Express	A Palestinian was shot, beaten and tied to an Israeli army jeep.	Unknown	Palestinians	The sentence correctly identifies the victim and the lack of mention of who was the perpetrator.	Both correctly identified.

Table 4.5: Evaluation of test-set sentences processed by the agent-patient modelling pipeline

4.3 Time Analysis of Articles Published

Figures 4.1 and 4.2 display the temporal distribution of published articles across news outlets, charted monthly from October 2023 onward. An important thing to note is that the y-axis of all articles are differently scaled to their respective counts. It was not normalized to show the pattern more clearly. We can make the following observations from these charts:

- October 2023 - Almost all outlets start with a sharp peak in this month. (Exceptions are New York Post and USA Today, both of which peak during November 2023 instead.) This suggests that they extensively reported during the October 2023 attacks by Hamas on Israel. For most outlets, events after October 2023 register only a fraction of the published articles. However, we can note that both India and Indian Express do come close.
- There are event-driven bumps during the highlighted events but none of these upticks are sustained. In March and April 2024, Israel had announced its plans to siege Rafah. This decision was met with disapproval from President Biden. In April 2024, Iran attacked Israel and the influence of these events caused an uptick in several outlets.
- In May 2024, when Israel started its offensive on Rafah, several outlets are shown to have a decrease in the number of relevant articles published than the month before. Exceptions include Fox News, Indian Express, News18, Newsweek and USA Today.
- Several outlets show a local maxima during October 2024. This was when Israel began its siege of the Jabalia camp and Hamas leader Yahya Sinwar was killed in a refugee camp in Rafah.
- In February and March 2025, Israel and Hamas exchanged hostages but failed to agree to a ceasefire. This is another point where a majority of the outlets show an event-driven peak.
- DailyMail UK, The Independent UK: Both show the largest absolute volumes across the observed period. Both show a massive Oct-2023 spike, then persistent but volatile output with noticeable event-linked bumps.

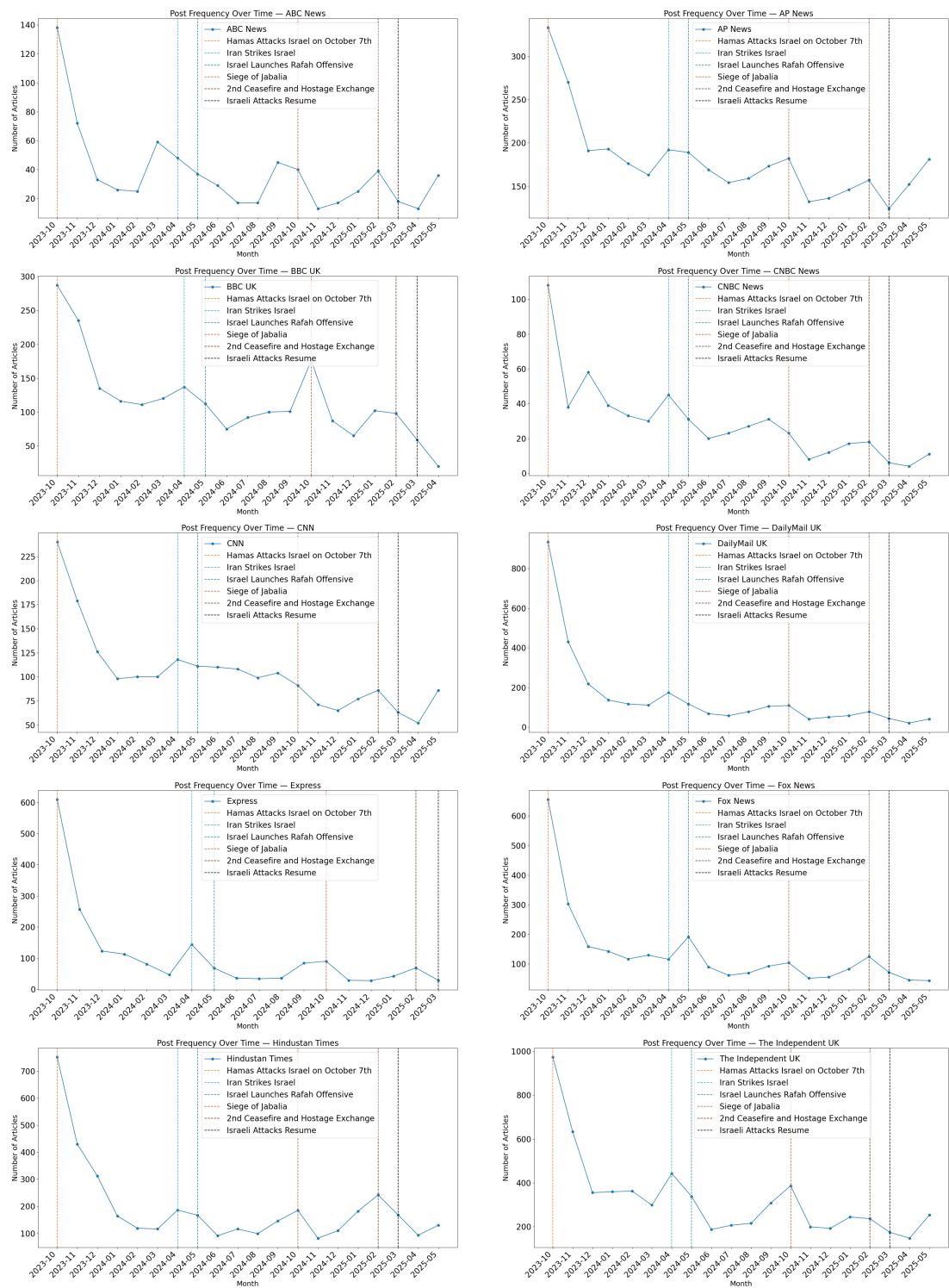


Figure 4.1: Number of articles published by each source every month from 7th Oct 2023 to 31st May 2025

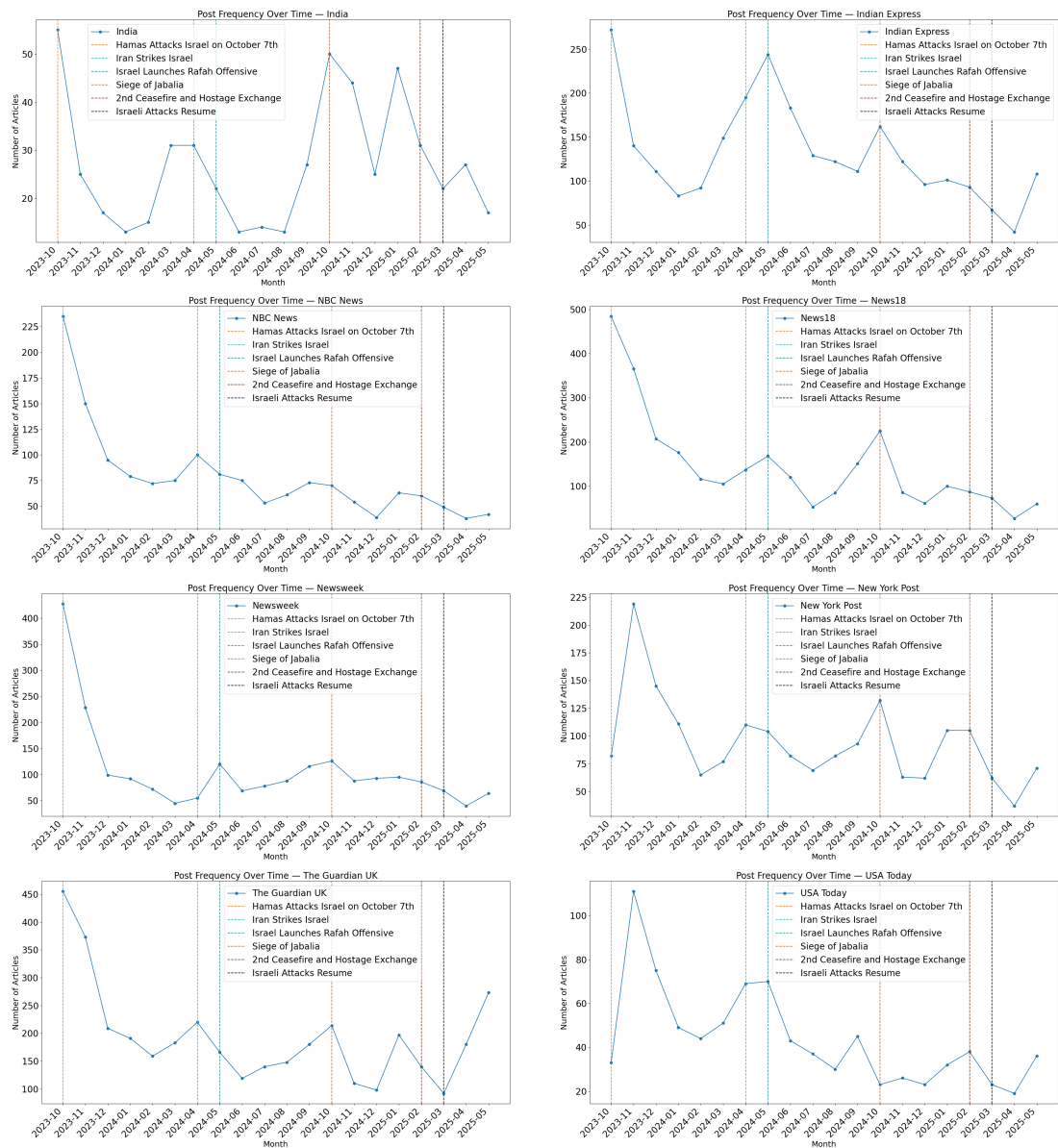


Figure 4.2: Number of articles published by each source every month from 7th Oct 2023 to 31st May 2025

- Fox News, New York Post: They show high early peaks and a spikier profile later, consistent with event-driven bursts.
- BBC UK, CNN, Guardian, AP News: The display a high initial interest, then stable, moderate monthly counts with clear but proportionate event bumps. AP (as a wire source) looks remarkably consistent post-peak.
- NBC News, USA Today: Lower end by 2025 with relatively flat tails. The volume response is modest.

- Hindustan Times, Indian Express, News18: There are large early surges, then lower baselines. News18 maintains a higher mid-period output than the other two.
- Express UK, Newsweek: They have moderate number of articles with clean, event-aligned upticks but nothing upto initial peak levels.

4.4 Analysing Violent Verbs in Sentences

4.4.1 Investigating Individual Sentences Processed by the Model

This section and Tables 4.6 and 4.7 discusses some of the sentences analysed by the model, the outputs and why the model is accurate or not from the unseen data.

Source	Sentence	Agent	Patient	Analysis
ABC News	3 Israeli hostages killed by IDF identified, were waving 'white cloth'	Israel	Israel	This is an example of cases where both the agent and patient are assigned the same side.
ABC News	Former ABC News journalist Husam al-Titi was killed in an Israeli airstrike in Gaza on Monday evening local time, the night the ceasefire between Israel and Hamas ended, according to al-Titi's family	Israel	Unknown	In sentences like these, where proper nouns are used, it is difficult to determine if the victim of the violent act can be referred to as part of a particular side. This leads to them being categorized as Unknown.
ABC News	Over the past 182 days, as Israel pursued its goal of destroying Hamas, a serious humanitarian crisis has unfolded in Gaza, with shortages of food, clean water and medicine, as well as a collapse of the health care system.	Israel	Palestinians	For the verb 'destroying', the sides are assigned accurately. However, with the way the sentence is structured, the latter half of the sentence, even though it denotes a negative event (humanitarian crisis), there is no violent verb. Cases like these, fail to be represented in the model.
AP News	The war in Gaza is wiping out entire Palestinian families, one branch at a time.	Palestinians	Palestinians	This sentence does not explicitly mention Israel at all, softening the responsibility. The blame is assigned to 'the war in Gaza' which makes the model process Palestinians as the agent and victim of violence.

Table 4.6: Evaluation of sentences and agent-patient assignment from unseen data

After performing SRL, Named Entity Recognition and coreference to each sentence, a CSV file was generated for each source domain, containing sentences and their violent verbs, along with identified agents and patients. Categories Unknown and Neither were

Source	Sentence	Agent	Patient	Analysis
AP News	Ruby Chen's son, Itay, was killed in the Hamas attack on Oct. 7 .	Palestinians	Unknown	The absence of any keyword in the patient results in the victim being labelled as Unknown.
BBC UK	An Israeli air strike killed 10 Palestinians in the north of the occupied West Bank on Wednesday night, the Palestinian health ministry says.	Israel	Palestinians	This is correctly classified as a case where Israel was the perpetrator and Palestinians were the victims.
CNBC	Hamas leader Ismail Haniyeh was assassinated in Tehran, Iran's paramilitary Revolutionary Guard said early Wednesday, and Hamas blamed Israel for the attack.	Palestinians	Palestinians	This sentence incorrectly assigns the blame to Palestinians. The article sentence itself does not confirm an agent of blame, Israel is only said to be 'blamed' for the attack. The model thus cannot recognize the agent.
AP News	Violence in the West Bank The Palestinian Health Ministry in the occupied West Bank said one Palestinian-American teen was killed and two others were injured — one in critical condition — and asserted that Israeli settlers had shot them.	Unknown	Palestinians	For the verb 'killed', the agent is unknown because the blame is only 'asserted' by the victims. For the verb 'shot', however, the agent is said to be Israeli.

Table 4.7: Evaluation of sentences and agent-patient assignment from unseen data

merged. For each source, their top 10 most-occurring violent verbs were calculated (4.8). All these verbs were unionized and their lemmas were then used to create the charts in Figures 4.3 and 4.4

Verbs	Root Lemma
'injured'	injure
'wipe'	wipe
'killed', 'killing', 'kill', 'kills',	kill
'seizing', 'seize', 'seized'	seize
'bombed'	bomb
'executed'	execute
'died', 'dying'	die
'shot'	shoot
'destroy', 'destroyed',	destroy
'wounded'	wound
'attack', 'attacked'	attack
'murdered'	murder

Table 4.8: Evaluation of enhanced agent-patient detection using SRL and Coreference

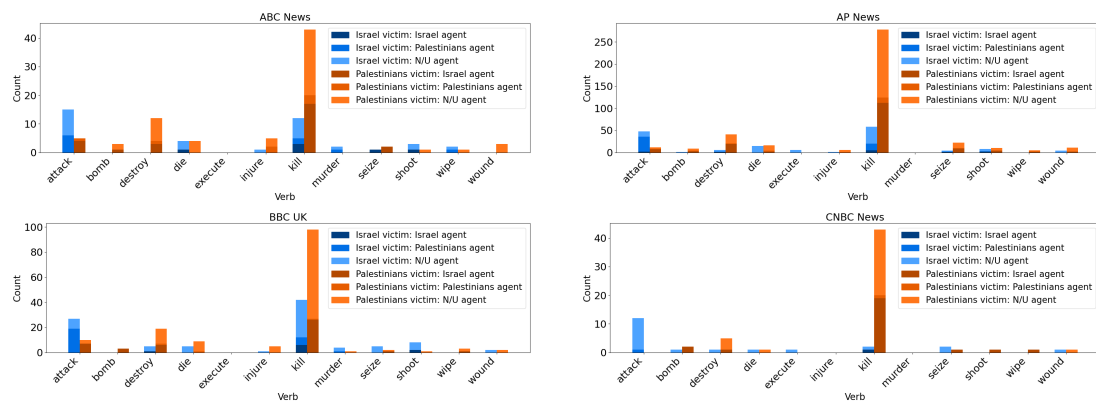


Figure 4.3: Top 10 Violent Verbs used in ABC News, AP News, BBC UK and CNBC along with the identification of Israeli and Palestinian agents and patients.

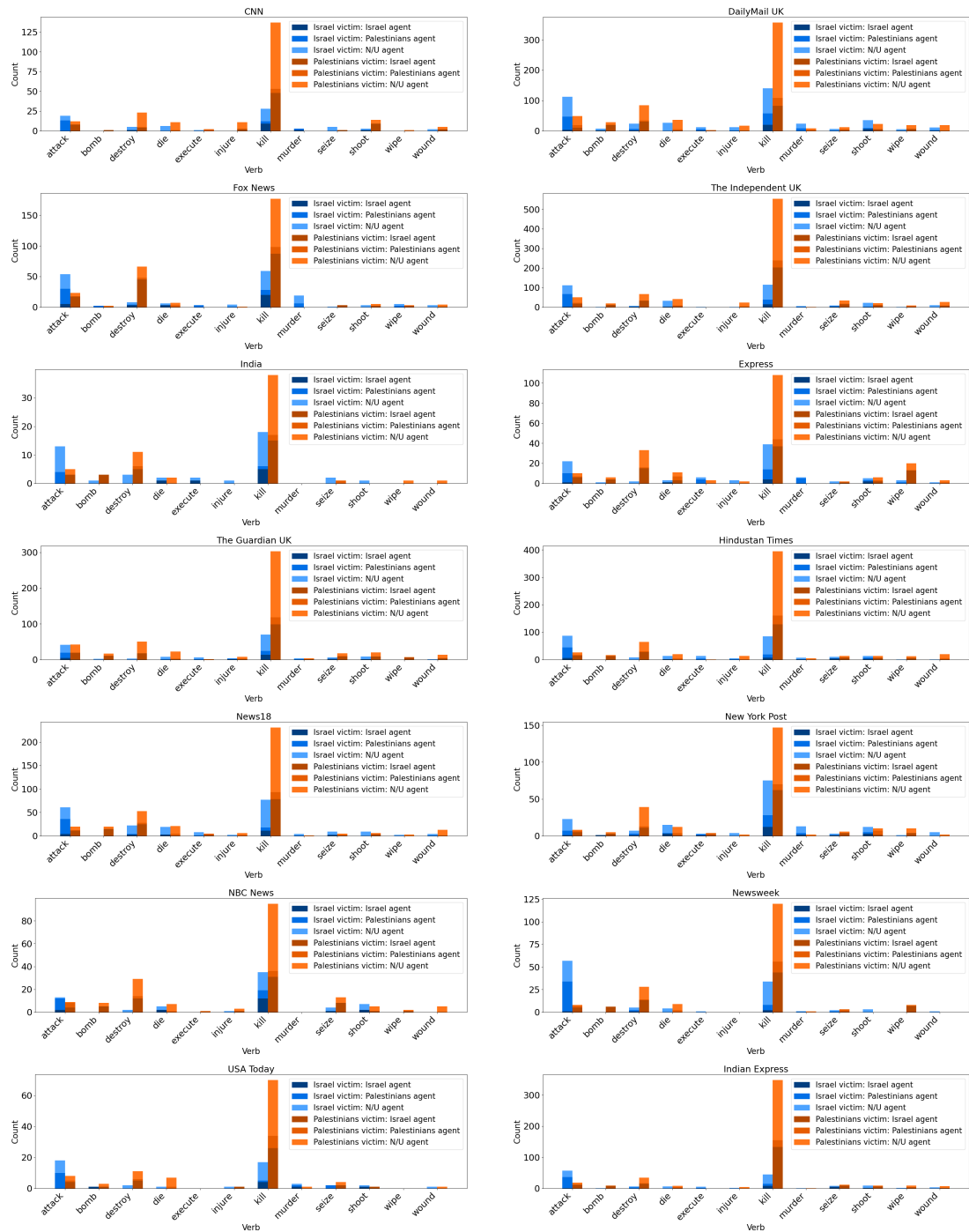


Figure 4.4: Top 10 Violent Verb Lemmas used in , CNN, DailyMail UK, Fox News, The Independent UK, India, Express UK, The Guardian, Hindustan Times, News18, NYPost, NBC News, Newsweek, USA Today along with the identification of Israeli and Palestinian agents and patients.

The analysis revealed that across outlets, kill was by far the most dominant verb, and Palestinians were overwhelmingly represented as the victims of violence. This pattern is not surprising, given that casualty statistics consistently demonstrate a disproportionate number of Palestinian deaths compared to Israeli deaths since October 2023. Thus, this quantitative asymmetry in victim representation cannot be attributed, in itself, to evidence of bias. It simply addresses the empirical reality of the conflict.

Agent attribution varies, but in many cases, the agent is labeled as “N/U” (Unknown/Unclassified), especially in the context of Palestinian victims. This reflects systematic linguistic choices (passive constructions and agent omission) or reporting conventions that obscure attribution. The agent responsible for violence is often left unspecified. In cases of Israeli victims, violence is tied to Palestinian agency. In contrast, for Palestinian victims, responsibility is often obscured. This suggests that while Palestinians appear more often as victims, their victimhood is linguistically framed in ways that diffuse or soften Israeli responsibility.

Regional differences in coverage further underscore this point. U.S. outlets, such as CNN, NBC, and USA Today, showed a greater reliance on unspecified agents when Palestinians were killed. Indexing Theory [9] states “the range of debate on public affairs appearing in the news is indexed to the range of debate present in mainstream government discourse.” Since the United States are an ally to Israel, the news coverage tends to reflect this. U.K. outlets, including the BBC, Guardian, and Independent, displayed higher coverage intensity and broader verb repertoires, more frequently attributing Palestinian deaths directly to Israeli actions. Indian outlets, such as Hindustan Times and News18, showed similarly high coverage intensity but were even more explicit in naming agents on both sides, reducing ambiguity and suggesting a journalistic style less constrained by diplomatic alignment.

These findings resonate with Entman’s (1993) theory of framing [13], which highlights how linguistic choices foreground certain actors over others. By naming Palestinian agency in Israeli deaths while often obscuring Israeli agency in Palestinian deaths, U.S. outlets construct an asymmetrical framing of responsibility. U.K. and Indian outlets, by contrast, more frequently present violence in explicitly attributed terms, producing discourses that foreground accountability. The variation across regions suggests that linguistic framing is not merely a product of journalistic convention but is also shaped by geopolitical contexts and editorial traditions.

While Palestinians are more often depicted as victims, Israelis are more often depicted as victims of named aggression. This imbalance points to asymmetry in how agency and culpability are assigned. The word choice between 'kill' and 'attack' particularly reflects this. Most of the cases with Palestinians as victims have been portrayed using 'kill'. On the other hand, 'attack' appears much more often in the case of Israeli victims with Palestinian agents. This constructs Palestinians as not only as victims of violence but also perpetrators of aggressive acts. Verb selection and agent attribution affect readers' perceptions of legitimacy and responsibility. This aligns with a narrative framing that portrays Palestinians as aggressors and Israelis as victims, reinforcing a specific power dynamic and moral judgment.

If we analyse the patterns observed in each outlet, Fox News placed greater emphasis on Israeli victimhood than other U.S. outlets, consistently naming Palestinian agents when Israelis were attacked or killed. CNN and AP displayed more verb variety but still demonstrated asymmetry in agency attribution. The Guardian and Independent were more explicit in linking Israeli actions to Palestinian deaths, while the BBC adopted a somewhat more balanced descriptive range. Indian outlets showed the highest intensity and the lowest ambiguity in agent naming, suggesting less constraint by geopolitical alignment and a preference for explicit reporting.

Finally, a question arises of cases where the patients and agents are the same. The following section explains a variety of sentences that the model processed.

4.5 Bias Summary

To investigate the difference in representation of Palestinians and Israelis in reporting, we focus on specific patient scenarios. Instances in which Palestinians were positioned as the patient in and conversely, instances in which Israelis were depicted as the patient. This way, we were able to isolate the contexts in which each group was portrayed as the victim of violent acts with the blame being attributed to either Unknown category, or the other respective side.

To account for the substantial variation in the number of articles produced by different media sources, the analysis was normalized by calculating the percentage representation of these scenarios within each source. This approach allows for a fair comparison across outlets with differing publication volumes, ensuring that the resulting visualization

(Figure 4.5) reflects relative emphasis rather than absolute counts.

We observe that The Guardian has the highest portrayal of Palestinian victims. Fox News displays a near-equal distribution of victimhood between Palestinians and Israelis. This is particularly noteworthy given that, according to independent conflict monitoring data, Palestinian casualties were significantly higher than Israeli casualties during the study period. The equal representation of Israeli and Palestinian victimhood in Fox News coverage, despite this imbalance, suggests a skewed framing that may obscure the asymmetry of suffering on the ground. This can imply underlying editorial biases, under-representing Palestinians and victims of violent acts.

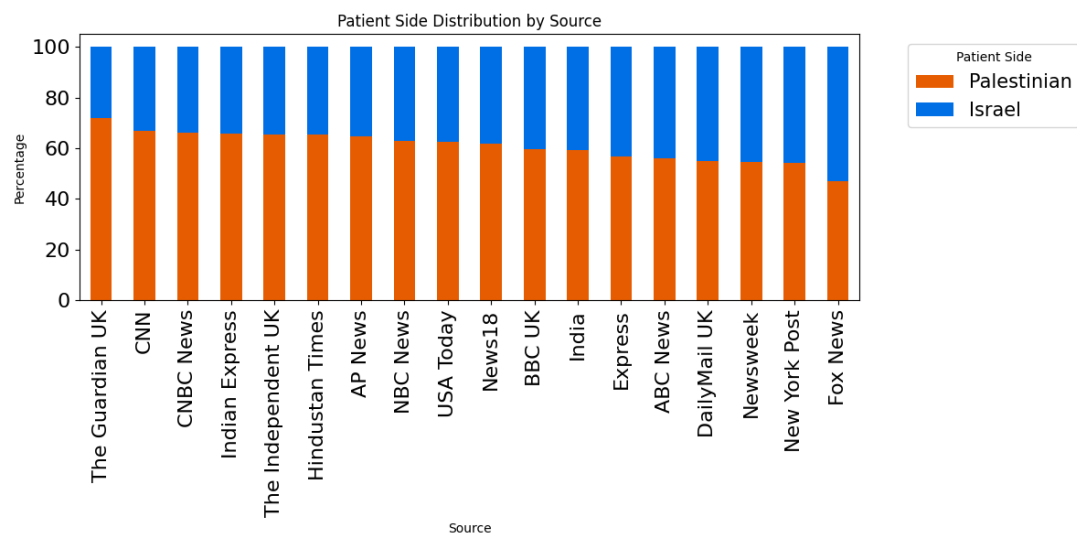


Figure 4.5: Representation of Palestinian and Israeli victims across different sources

Chapter 5

Conclusion

This study builds an extensive corpus of articles published about the Palestine-Israel conflict and builds a semi-supervised classification model that filters relevant articles.

Addressing the research questions that the study follows:

How are agents and patients of violence linguistically encoded across different news sources covering the Israeli-Palestinian conflict? There is an empirical difference observed in the representation of Palestinian victims, due to the counts of casualty. However, linguistic choices are often made by news outlets in how they present stories with Palestinian victims and Israeli victims. There was also a regional difference observed. Outlets from the United States proved to be more likely to remove the agent in cases with Palestinian victims, failing to correctly portray the actor in these cases. The choice of verbs used also show that the narrative of aggression is pushed towards one side more than the other, even in case of violent acts only.

How effectively can computational methods such as semantic role labeling combined with verb classification capture and quantify these framing phenomena? Semantic Role Labelling alone is not enough in itself to be able to recognise agents and patients for verbs correctly. However, combining it with coreference does improve the model's capacity to identify actors in complexly worded sentences. There are further limitations it faces like confusion regarding very generic terms 'militants', 'troops' and similarly, it is unable to classify Proper Nouns due to lack of information.

This study therefore worked on building an extensive corpus of articles published by eight different news media websites from around the world. It then built a semi-supervised classifier for finding articles that specifically talk about the conflict itself and official comments made regarding the conflict. The corpus was used to perform two different analysis:

1. Time analysis on the frequency of articles published since the October 7th attacks and
2. Analysis of the reporting of violent acts and the linguistic choices made and the framing of blame.

5.1 Limitations

The following are the limitations of the methods used in the study:

- **Small Training Sample used for Semi-Supervised Classification** - There were only 200 labelled articles on which the initial dataset is trained for relevance-based classification.
- **Small Dataset for Evaluation of Agent-Patient Detection Model** - Only 60 randomised samples were used to check the performance of the model. This is not representative of the sentences.
- **Model's Limitations** - Using techniques like NER, SRL and Coreference can work for simple sentences, but with long and complicated sentences, it becomes harder for them to track the actors. The model also depends on a pre-defined list of terms that would be used to describe different sides. If there are names of people that it does not recognise, they are classified as N. Similarly, for locations, if a new area appears in only some articles with no reference to generic terms like 'Gaza', 'Tel-Aviv', the model will be unable to correctly recognise the agents and patients. Similarly, the model was limited in its ability to handle implicit or metaphorical spatial references (e.g., "on the border," "inside occupied territory"). Finally, coreference remains a difficult task. For long sentences, it becomes harder for the model to track entities.

5.2 Future Work

This study creates a model to detect agents and patients in media reporting of acts of violence. It can be used to create a well-rounded framework that can analyse different bias metrics in the Palestine-Israel conflict and beyond.

- Improve agent-patient identification by integrating advanced coreference resolution models that handle implicit references more effectively, reducing ambiguous or missing role assignments.
- Analyse the linguistic choices made over time to see if they change over time. This could give more insight into how the editorial choices for outlets evolve over time and to understand the dynamics of media narratives and agenda shifts. For example, if certain violent verbs are heavily attributed to a specific keyword, instead of the general side.
- Employ transformer-based language models that incorporate broader discourse context to detect framing and bias more accurately, surpassing sentence-level analysis.
- Conduct user studies to correlate computational bias measures with audience perception, enhancing the relevance and application of automated bias detection.
- Build a well rounded media-bias framework that takes into account different metrics including the framing of agents and patients and words used to describe different sides.

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Appendix A

First appendix

All project materials can be found in SMASH server at:

`/mnt/raid1/MSCs/AY2425/kjain/msc-diss.`

Supplementary project materials have been submitted along with this report.